Reverse of differentiation

1 Consider the function x^4.

a Calculate $\frac{d}{dx}\left(x^4\right)$.

b Hence find $\int 4x^3 dx$.

2 Consider the function x^6.

a Calculate $\frac{d}{dx}\left(x^6\right)$.

b Hence find $\int 24x^5 dx$.

3 Consider the function $x^5 + x^4$.

a Calculate $\frac{d}{dx}\left(x^5 + x^4\right)$.

b Find $\int \left(35x^4 + 28x^3\right) dx$.

4 Consider the function x^{-4}.

a Calculate $\frac{d}{dx}\left(x^{-4}\right)$.

b Hence find $\int 16x^{-5} dx$.

5 Consider the function $x^8 + x^6$.

a Calculate $\frac{d}{dx}\left(x^8 + x^6\right)$.

b Find $\int \left(4x^7 + 3x^5\right) dx$.

6 Consider the function $\sqrt[5]{x^6}$.

a Calculate $\frac{d}{dx}\left(\sqrt[5]{x^6}\right)$.

b Hence find $\int 6\sqrt[5]{x} dx$.

Anti-differentiation

7 Consider the gradient function $f'(x) = 2$.

a Write the general form of the antiderivative function, $f(x)$.

b The gradient function $f'(x)$, is a constant function. What type of function is the antiderivative function, $f(x)$.

c If the form of a constant gradient function is $f'(x) = m$, write the general form of the antiderivative function.

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8 Consider the gradient function $f'(x) = -8x$.

a Write the general form of the antiderivative function, $f(x)$.

b The gradient function $f'(x)$, is a linear function. What type of function is the antiderivative function, $f(x)$.

c If the form of a linear gradient function is $f'(x) = ax$, write the general form of the antiderivative function.

9 Consider the gradient function $f'(x) = 4x + 3$.

a Write the general form of the antiderivative function, $f(x)$.

b The gradient function $f'(x)$, is a linear function. What type of function is the antiderivative function, $f(x)$.

c If the form of a linear gradient function is $f'(x) = ax + b$, write the general form of the antiderivative function.

10 Consider the gradient function $f'(x) = 7x^2$.

a Write the general form of the antiderivative function, $f(x)$.

b The gradient function $f'(x)$, is a quadratic function. What type of function is the antiderivative function, $f(x)$.

c If the form of a quadratic gradient function is $f'(x) = ax^2$, write the general form of the antiderivative function.

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11 Find the primitive function of the following:

a $\frac{dy}{dx} = 9$

b $\frac{dy}{dx} = 8x$

c $\frac{dy}{dx} = 10x + 7$

d $\frac{dy}{dx} = 9x^2$

e $\frac{dy}{dx} = 9x^2 + 4x - 6$

f $\frac{dy}{dx} = x^5$

g $\frac{dy}{dx} = 15x^4 + 16x^3$

h $\frac{dy}{dx} = \frac{x^6}{4} + \frac{x^2}{3}$

i $\frac{dy}{dx} = x^{-6}$

j $\frac{dy}{dx} = \frac{15}{x^6}$

k $\frac{dy}{dx} = \frac{10}{x^6} - \frac{9}{x^4}$

l $\frac{dy}{dx} = 4x^{\frac{2}{5}} + 3x^{\frac{4}{7}}$

m $\frac{dy}{dx} = x^{-\frac{3}{7}} + x^{-\frac{2}{5}}$

n $\frac{dy}{dx} = 8x^3 + 3x^{\frac{5}{3}} - 3$

o $\frac{dy}{dx} = \frac{x^3 + 4}{x^3}$

p $\frac{dy}{dx} = x^2\left(10x^2 - 9x\right)$

q $\frac{dy}{dx} = (5x - 2)(3x - 4)$

r $\frac{dy}{dx} = (x + 4)(x + 6)$

s $\frac{dy}{dx} = \sqrt{x}$

t $\frac{dy}{dx} = 18\sqrt{x}$

u $\frac{dy}{dx} = x\sqrt{x}$

v $\frac{dy}{dx} = \frac{6}{\sqrt{x}}$

Applications

12 For each of the following gradient functions:

i State what type of function the antiderivative, $f(x)$, is.

ii Find $f(x)$.

iii Sketch a possible graph for the antiderivative $f(x)$.

a $f'(x) = 6$

b $f'(x) = 8x + 8$

c $f'(x) = 7x^2$

d $f'(x) = -5x(x + 2)$

13 Consider the gradient function $f'(x) = \frac{3}{x^2}$.

a Find $f(x)$.

b Sketch a possible graph of the antiderivative $f(x)$.

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14 The gradient function, $f'(x)$, has only one x-intercept at $(-4, 0)$, a y-intercept at $(0, -3)$ and a constant gradient.

- **a** Find $f'(x)$.
- **b** Find $f(x)$.
- **c** Sketch a possible graph of the antiderivative $f(x)$.

Find value of C

15 Consider the equation $\frac{dy}{dx} = 4x + 7$.

- **a** Find a general equation for y.
- **b** Find the equation of y, if the curve passes through the point $(3, 41)$.

16 Consider the equation $\frac{dy}{dx} = 9x^2 - 10x + 2$.

- **a** Find a general equation for y.
- **b** Find the equation of y, if the curve passes through the point $(2, 13)$.

17 A family of curves has a gradient function $y' = 15x^2 + 7$.

- **a** Find the equation of y for the family of curves.
- **b** Find the equation of the curve that passes through the point $(2, 59)$.

18 Consider the gradient function $\frac{dy}{dx} = 10x^4 + 20x^3 + 6x^2 + 6x + 9$.

- **a** Find an equation for y.
- **b** Find the equation of y, if the curve passes through the point $(-3, -133)$.

19 Consider the gradient function $\frac{dy}{dx} = 9x^{\frac{2}{3}}$.

- **a** Find an equation for y.
- **b** Find the equation of y, if the curve passes through the point $\left(8, \frac{889}{5}\right)$.

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20 Find the equation of a curve p in terms of t, given the following:

- $\frac{dp}{dt} = 6t - 5$
- when $t = 3$, $\frac{dp}{dt} = 13$ and $p = 15$

21 Find the equation of a curve y in terms of x, given the following:

- $y'' = 0$.
- y contains the points $(-1, -1)$ and $(0, 2)$.

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